06/06/2026

Code-P

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot No. 13, Sector-18,
Udyog Vihar, Gurugram, Haryana - 122015, ***Ph.***+91-1244168300

MM : 720 **Project Phoenix for Re-NEET-2026_CST-06P** **Time : 180 Min.**

Topics Covered:
Complete Syllabus of Class XI & XII

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are three parts in the question paper consisting of Physics, Chemistry and Biology which have 45 questions each in Physics & Chemistry and 90 questions in Biology.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. A potential difference of $V = (20 \pm 0.1)$V when applied across a resistance gives a current of (2.5 ± 0.05) A. The resistance of the wire with limit of error will be expressed as

(1) $(8 \pm 0.5)\ \Omega$

(2) $(8 \pm 1)\ \Omega$

(3) $(8 \pm 0.2)\ \Omega$

(4) $(8 \pm 1.5)\ \Omega$

2. A force $\vec{F} = 8\left(y\hat{i} + x\hat{j}\right)$ N, where x and y co-ordinates are measured in metres, acts on a particle moving in the xy-plane. The particle is taken from (2, 2) to $(\sqrt{2}, 0)$. Find the work done by the force $\vec{F}$ on the particle.

(1) –16 J

(2) –32 J

(3) –8 J

(4) –48 J

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3. Given are two statements:
Statement I: Area under velocity-time graph gives the distance travelled.
Statement II: For one-dimensional motion, a constant speed in an interval must have non-zero acceleration in that interval.
In the light of above statements, choose the most appropriate answer from the following

(1) Both statements I and II are correct

(2) Both statements I and II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

4. Rain is falling vertically with velocity 10 m/s and a man is moving horizontally with velocity 6 m/s. The angle with which umbrella should be hold (with vertical) by man to avoid getting wet is

(1) $\tan^{-1}\left(\frac{5}{3}\right)$

(2) $\tan^{-1}\left(\frac{2}{3}\right)$

(3) $\tan^{-1}\left(\frac{3}{5}\right)$

(4) $\tan^{-1}\left(\frac{3}{2}\right)$

5. A river 400 m wide is flowing at a rate of 2.0 m/s. A boat is sailing at a velocity of 10 m/s with respect to the water, in a direction perpendicular to the river. The time taken by the boat to reach the opposite bank is

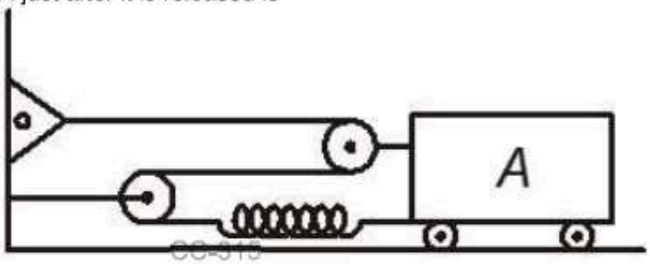
(1) 80 s

(2) 200 s

(3) 40 s

(4) 20 s

6. Block *A*, with mass m_A , is pulled to the right a distance *x* from the position in which the spring is relaxed. It is then released from the rest. Assuming pulley, strings and wheels are ideal. The spring constant is *k*. The acceleration of block *A* just after it is released is



(1) $\frac{3kx}{m_A}$

(2) $\frac{2kx}{m_A}$

(3) $\frac{3}{2}\frac{kx}{m_A}$

(4) $\frac{9kx}{m_A}$

7. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion(A): A moving body may have zero distance covered during a time interval.
Reason(R): If a body returns to its original position then displacement covered by the body becomes zero.
In the light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

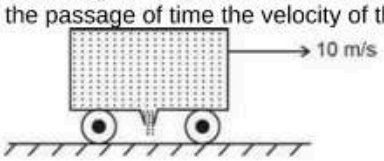
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

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8. A trolley filled with sand is moving on a smooth horizontal surface with a velocity of 10 m/s, as shown in the figure. Sand begins to spill out from a small hole at the bottom. With the passage of time the velocity of the trolley will



(1) Will increase

(2) Will decrease

(3) Remain same

(4) None of these

9. A uniform rod of mass *m* and length *l* is pivoted at one end and is rotated with constant angular velocity ω in a horizontal plane. Tension in the rod at a distance $\frac{\sqrt{3}l}{2}$ from the pivoted end is (Assume gravity free space)

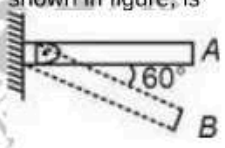
(1) $\frac{3}{4}m\omega^2\ell$

(2) $\frac{m\omega^2\ell}{2}$

(3) $\frac{m\omega^2\ell}{4}$

(4) $\frac{m\omega^2\ell}{8}$

10. A rod of mass 4 kg and length 3 m is released from position *A*. The angular acceleration of the rod at position *B* as shown in figure, is



(1) 5 rad/s^2

(2) 2.5 rad/s^2

(3) 3 rad/s^2

(4) 3.5 rad/s^2

11. The moment of inertia of a uniform rectangular plate of mass *M*, with edges of length *l* and *b*, about its axis passing through the centre and perpendicular to the plate, will be

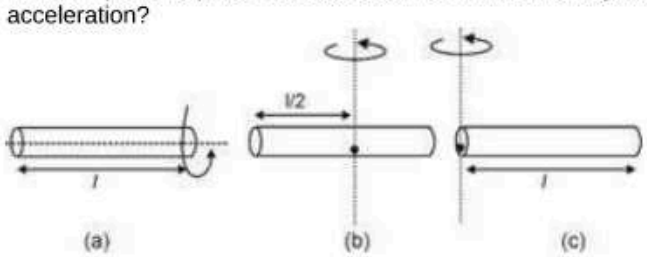
(1) $\frac{Ml^2}{12}$

(2) $\frac{Ml^2}{3}$

(3) $\frac{M(l^2+b^2)}{12}$

(4) $\frac{Mb^2}{3}$

12. A rod of length *l* and mass *m* is rotated about three different axes as shown in figure. In which of the following cases, more torque is required to rotate each rod with same angular acceleration?



(1) In case (a)

(2) In case (b)

(3) In case (c)

(4) Both (1) and (2)

13. Two balls each of mass 2 kg (one at rest) undergo oblique collision. If collision is perfectly elastic, then the angle between their velocities after collision is

(1) 30°

(2) 60°

(3) 45°

(4) 90°

14. Two positive point charges of 9 μC & 6 μC are 10 cm apart. The work done in bringing them to a separation of 4 cm is

(1) 5.25 J

(2) 6.29 J

(3) 7.29 J

(4) 8.25 J

15. A satellite of mass *m* is orbiting the earth at an altitude $\frac{R}{3}$, where *R* is radius of earth. Extra energy given to the satellite to increase its altitude to *R*, will be(*g* is the acceleration due to gravity at the surface of earth)

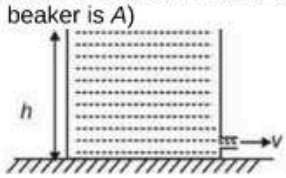
(1) *mgR*

(2) $\frac{mgR}{2}$

(3) $\frac{mgR}{8}$

(4) $\frac{mgR}{3}$

16. A large beaker of height *h* is completely filled with water of mass *m*. A small hole of area *a* is made at the bottom of the beaker as shown. Assuming mass of the beaker to be negligible, the minimum value of coefficient of friction such that the beaker remains at rest, is (Area of cross-section of beaker is *A*)



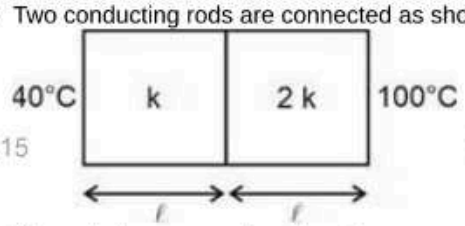
(1) 2

(2) $\frac{3}{2}$

(3) $\frac{2A}{a}$

(4) $\frac{2a}{A}$

17. Two conducting rods are connected as shown



The rods have same length and same cross-section area. In the steady state the temperature of the interface will be (in °C)

(1) 70°C

(2) 60°C

(3) 80°C

(4) 90°C

18. A body at temperature 127°C radiates energy at a rate equal to *E*. If its temperature is increased by 400°C and surface area is reduced to half of the initial value, then the rate of radiation will be

(1) *E*

(2) 4*E*

(3) 6*E*

(4) 8*E*

19. Given below are three statements:

Statement A: In a rigid body, angular velocity of any point with respect to any other point on the body is constant.

Statement B: Moment of Inertia of a rigid body does not depend on angular momentum.

Statement C: Theorem of perpendicular axis is applicable for planar bodies.

In the light of the above statements, choose the correct option.

(1) Only statement A is true

(2) Only statement B is true

(3) Both the statements A and B are false

(4) All the statements are true

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20. An ideal monatomic gas is taken through a process $\Delta Q = 3\Delta U$, the molar heat capacity of the gas for this process will be (where symbols have their usual meaning)

(1) 4.5*R*

(2) 3.5*R*

(3) 4*R*

(4) 5*R*

21. Pressure *P*, volume *V* and Temperature *T* of a certain gas are related by, $P = \frac{\alpha T^2}{V}$,where α is a constant. The work done by the gas when temperature changes from T_0 to $3T_0$ while pressure remains constant, is

(1) $4\alpha T_0^2$

(2) $8\alpha T_0^2$

(3) $2\alpha T_0^3$

(4) $4\alpha T_0^3$

22. The equation of the transverse wave travelling in a rope is given by $y = 20\sin(x - 200t)$ where *y* and *x* are in cm and *t* in second. The maximum particle speed in rope is

(1) 20 m/s

(2) 40 m/s

(3) 10 m/s

(4) 30 m/s

23. If the refractive index of a material of equilateral prism is $\sqrt{3}$ then angle of minimum deviation of the prism is

(1) 30°

(2) 45°

(3) 60°

(4) 75°

24. A particle is executing S.H.M with amplitude *A*. Its displacement from the mean position when its kinetic energy is three fourth of its total energy is

(1) $\frac{A}{4}$

(2) $\frac{A}{\sqrt{2}}$

(3) $\frac{A}{2}$

(4) $\frac{\sqrt{3}A}{2}$

25. Match column I and column II and choose the correct option.

Column I	Column II
(a) For translational equilibrium	(p) $\Sigma\vec{\tau} = 0$
(b) For rotational equilibrium	(q) $\Sigma\vec{F} = 0$
(c) Torque is required to produce	(r) $\frac{7}{5}MR^2$
(d) Moment of inertia of solid sphere (*M*, *R*) about tangent	(s) Angular acceleration

(1) (a)-(r), (b)-(p), (c)-(q), (d)-(s)

(2) (a)-(q), (b)-(p), (c)-(s), (d)-(r)

(3) (a)-(q), (b)-(s), (c)-(p), (d)-(r)

(4) (b)-(r), (b)-(q), (c)-(s), (d)-(p)

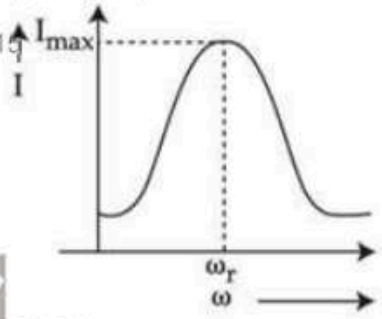
26. For a series LCR circuit, *I* vs ω curve is shown :

(a) To the left of ω_r , the circuit is mainly capacitive.

(b) To the left of ω_r , the circuit is mainly inductive.

(c) At ω_r , impedance of the circuit is equal to the resistance of the circuit.

(d) At ω_r , impedance of the circuit is 0.



Choose the most appropriate answer from the options given below

(1) (a) and (d) only

(2) (b) and (d) only

(3) (a) and (c) only

(4) (b) and (c) only

27. A transformer has 1000 turns in primary coil and 3000 turns in secondary coil. If an AC supply of $V = 100\sin(314\pi t)$ is applied across primary coil, then peak value of emf across secondary coil will be

(1) 50 V

(2) 200 V

(3) 33.3 V

(4) 300 V

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28. A steady current 'I' flows in a small square loop of wire of side *L* in a horizontal plane. The loop is now folded about its middle such that half of it lies in a vertical plane. Let $\vec{\mu}_1$ and $\vec{\mu}_2$ respectively denote the magnetic moments of the current loop before and after folding. Then

(1) $\vec{\mu}_2 = 0$

(2) $\frac{|\vec{\mu}_1|}{|\vec{\mu}_2|} = \sqrt{2}$

(3) $\frac{|\vec{\mu}_1|}{|\vec{\mu}_2|} = \frac{1}{\sqrt{2}}$

(4) $\vec{\mu}_1$ and $\vec{\mu}_2$ are in the same direction

29. Displacement current exists within the gap between the plates of capacitor when the electric field between its plates

(1) Increases with time

(2) Decreases with time

(3) Remains constant

(4) Both (1) and (2)

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30. A conducting rod of length *l* is moved with a constant velocity $\vec{V}$ in a magnetic field ($\vec{B}$). A potential difference may appear across the two ends if

(1) $\vec{V} \parallel \vec{l}$

(2) $\vec{V} \parallel \vec{B}$

(3) $\vec{l} \parallel \vec{B}$

(4) $\vec{V} \perp \vec{B}$

31. Eight identical spherical liquid drops each charged to potential of 2 volt. They coalesce to make one big spherical liquid drop, whose potential is

(1) $\frac{1}{4}$ V

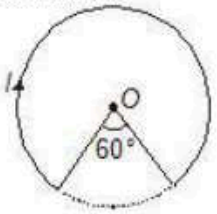
(2) 16 V

(3) 8 V

(4) 4 V

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32. The magnetic induction at the centre *O*, if the wire carries current *I*, is



(1) $\frac{5\mu_0 I}{12r}$

(2) $\frac{\mu_0 I}{2r}$

(3) $\frac{\mu_0 I}{2\pi r}$

(4) $\frac{\mu_0 I}{12r}$

33. A bubble in glass slab (μ = 1.5) when viewed from one side appears at 6 cm and at 4 cm when viewed from other side, then thickness of slab is

(1) 10 cm

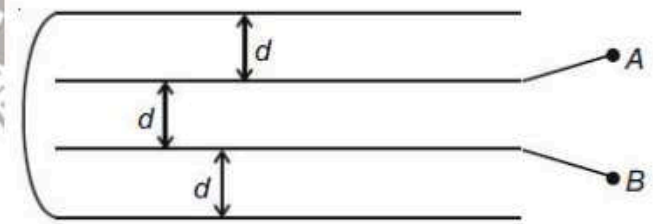
(2) 15 cm

(3) 12 cm

(4) 6.67 cm

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34. Four identical metal plates are located in air at a distance *d* from one another. If the area of each plate is *A*, then the equivalent capacitance between *A* and *B* is



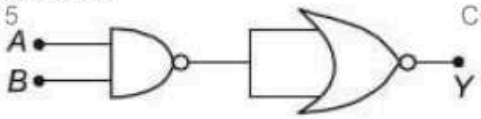
(1) $\frac{\varepsilon_0 A}{2d}$

(2) $\frac{3}{2}\frac{\varepsilon_0 A}{d}$

(3) $\frac{3\varepsilon_0 A}{d}$

(4) $\frac{\varepsilon_0 A}{d}$

35. The logic gate as shown in the figure performs the logic function of



(1) NAND gate

(2) NOR gate

(3) AND gate

(4) OR gate

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36. The geometrical path of ray of light in a medium of refractive index 2 is 8 m. The optical path is

(1) 16 m

(2) 4 m

(3) 2 m

(4) 1 m

37. In a Young's double slit experiment, a student observes 8 fringes in a certain segment of screen when a monochromatic light of 600 nm wavelength is used. If the wavelength of light is changed to 400 nm, then the number of fringes he would observe in the same region of the screen is

(1) 9

(2) 12

(3) 6

(4) 8

38. A screen is placed 100 cm from a single slit which is illuminated with 5000 Å light. If distance between the first and third minima in the diffraction pattern is 5 mm. The width of the slit is

(1) 2 mm

(2) 0.2 μm

(3) 2×10^{-4} m

(4) 4×10^{-5} m

39. The de-Broglie wavelength of a particle accelerating with 130 V potential is 1 Å. If the accelerating potential is increased to 520 V, then its de-Broglie wavelength will be

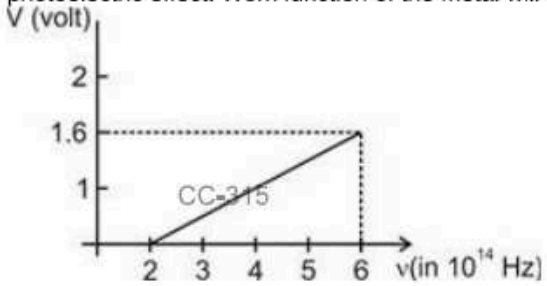
(1) 1 Å

(2) 0.25 Å

(3) 0.5 Å

(4) 2 Å

40. Figure given below represents the plot of stopping potential versus the frequency of light used in an experiment on photoelectric effect. Work function of the metal will be



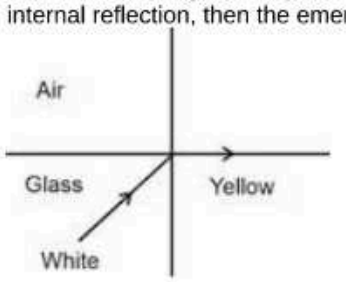
(1) 0.4 eV

(2) 0.2 eV

(3) 0.8 eV

(4) None of these

41. White light is incident on the interface of glass and air as shown in the figure. If yellow light just undergoes total internal reflection, then the emerging ray in air contains



(1) Violet and indigo only

(2) Violet, indigo, blue and green

(3) Orange only

(4) Orange and red only

42. A student measures the diameter of a small steel ball using a screw gauge of least count 0.001 cm. The main scale reading is 10 mm and with 25th divisions coincides with reference line. The correct diameter of the ball is (Number of circular scale division is 50)

(1) 1.35 cm

(2) 1.25 cm

(3) 1.025 cm

(4) 1.42 cm

43. Waves from two coherent sources with intensity $4I_0$ and $9I_0$ respectively, interfere at a point in medium. The minimum intensity would be

(1) $5I_0$

(2) $4I_0$

(3) $2I_0$

(4) I_0

44. The work function of metal is 1.2 eV. Light of wavelength 4000Å is incident on this metal surface. The velocity of emitted photo electrons will be

(1) 6.12×10^4 m/s

(2) 8.17×10^5 m/s

(3) 9×10^9 m/s

(4) 1×10^3 m/s

45. If the binding energy of deuteron $\left({}_1^2\mathrm{H}\right)$ is 1.15 MeV per nucleon and an alpha particle $\left({}_2^4\mathrm{He}\right)$ has a binding energy of 7.1 MeV per nucleon, then in the reaction ${}_1^2\mathrm{H} + {}_1^2\mathrm{H} \rightarrow {}_2^4\mathrm{He} + Q$. The energy Q released is

(1) 4.6 MeV

(2) 28.4 MeV

(3) 23.8 MeV

(4) 33 MeV

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CHEMISTRY

46. Match List I with List II and choose the correct option.

	List I (Amount of Substance)		List II (Number of oxygen atoms)
a.	18 g $C_6H_{12}O_6$	(i)	4 N_A
b.	44.8 L NO_2(g) at STP	(ii)	0.03 N_A
c.	1.225 g $KClO_3$	(iii)	0.5 N_A
d.	11.2 L CO(g) at STP	(iv)	0.6 N_A

(1) a(iv), b(i), c(ii), d(iii)

(2) a(i), b(ii), c(iii), d(iv)

(3) a(iv), b(iii), c(ii), d(i)

(4) a(ii), b(iii), c(i), d(iv)

47. 4.08 g of a mixture of BaO and unknown carbonate MCO_3 was heated strongly. The residue weighed 3.64g. This was dissolved in 100 mL of 1N HCl. The excess acid required 16 mL of 2.5 N NaOH solution for complete neutralisation. The metal M is (Atomic weight of Ba = 138)

(1) Ca

(2) Mg

(3) Sr

(4) Be

48. Given below are two statements: one is labelled as **Assertion(A)** and other is labelled as **Reason(R)**.
Assertion (A): Energy of 2*s* orbital of hydrogen atom is greater than that of 2*s* orbital of sodium.
Reason (R): Energies of the orbitals in the same subshell decrease with increase in the atomic number (Z_{eff}).
In the light of above statements, choose the correct answer from the options given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

49. The ground state energy of hydrogen atom is –13.6 eV. Consider an electronic state Ψ of Li^{2+} whose energy, azimuthal quantum number and magnetic quantum number are –7.65 eV, 1 and 0, respectively. Which of the following statement is true from the state ψ?
(a) It is a 4p state
(b) It has 2 radial nodes
(c) It has 2 angular nodes
The correct statement(s) is/are

(1) (c) only

(2) (a) only

(3) (a) and (b) only

(4) (b) and (c) only

50. The 1st, 2nd and the 3rd ionization enthalpies I_1, I_2 and I_3 of four atoms with atomic numbers n, n+1, n+2 and n+3, where n < 10, are tabulated below:

Atomic number	Ionization enthalpy (kJ/mol)		
	I_1	I_2	I_3
n	1682	3374	6050
n + 1	2082	3952	6122
n + 2	496	4562	6910
n + 3	739	1461	7733

The value of n is

(1) 7

(2) 8

(3) 9

(4) 6

51. Match the List I with List II

List I (Molecules)	List II (Shape)
a. XeF_4	(i) Trigonal bipyramidal
b. XeO_2F_2	(ii) Square planar
c. IF_5	(iii) Square pyramidal
d. PCl_5	(iv) See-saw

Choose the correct answer from the options given below.

(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(iv), c(iii), d(i)

(3) a(iv), b(ii), c(iii), d(i)

(4) a(iv), b(ii), c(i), d(iii)

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52. Correct order of bond length in O_2, O_2^+, O_2^- is

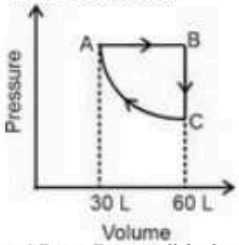
(1) $O_2^+ < O_2 < O_2^-$

(2) $O_2^- < O_2 < O_2^+$

(3) $O_2^- < O_2^+ < O_2$

(4) $O_2 < O_2^+ < O_2^-$

53. 3 moles of a monoatomic ideal gas undergo following transformations



• AB → Reversible isobaric expansion
• BC → Reversible isochoric process
• CA → Reversible isothermal compression at 27°C
The work done for the process C → A and the value of change in enthalpy for overall process respectively are

(1) 51.1 kJ and zero

(2) 51.1 kJ and 1.8 cal

(3) 5.17 kcal and 1.8 cal

(4) 1.24 kcal and zero

54. $\frac{1}{4}P_4(s) + \frac{3}{2}Cl_2(g) \rightarrow PCl_3(g) + x\ kJ$

$\frac{1}{4}P_4(s) + \frac{5}{2}Cl_2(g) \rightarrow PCl_5(g) + y\ kJ$

When phosphorous of purity 60% is allowed to react with insufficient chlorine, 40% of phosphorus is converted into 'PCl_3' and the remaining is converted in 'PCl_5'.
The heat generated (in kJ) when 0.31 kg of phosphorous is reacted with chlorine gas is

(1) x + y

(2) 3.6 (x + y)

(3) (3.6 x + 2.4 y)

(4) (2.4 x + 3.6 y)

55. The solubility (in mol/L) of Ag_2CrO_4 in a 0.2 M K_2CrO_4 solution is (Given : K_{sp} of Ag_2CrO_4 = 1.1×10^{-12} at 298 K and $\sqrt{5.5} = 2.34$)

(1) 1.17×10^{-5}

(2) 1.17×10^{-6}

(3) 1.4×10^{-8}

(4) 2.1×10^{-7}

56. Match solutions in List-I with their pH and nature in List-II.

List-I	List-II
(a) Aq CH_3COONa at 25° C	(i) pH > 7, basic
(b) Aq CH_3NH_3Cl at 25°C	(ii) pH < 7, acidic
(c) Aq NaCl at 25°C	(iii) pH < 7, neutral
(d) Water at 90°C	(iv) pH = 7; neutral

Correct option is

(1) (a)-(i), (b)-(iii), (c)-(ii), (d)-(iv)

(2) (a)-(i), (b)-(ii), (c)-(iv), (d)-(iii)

(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(4) (a)-(ii), (b)-(i), (c)-(iv), (d)-(iii)

57. The correct equivalent weight of Fe_2S_3 (Molar mass of Fe_2S_3 = M) in the reaction is

$Fe_2S_3 + O_2 \rightarrow Fe_2O_3 + SO_3$, is

(1) M/6

(2) M/12

(3) M/24

(4) M/2

58. Given below are two statements:
Statement I: Iodine can be used for titrating strong reducing agents like sodium thiosulphate.
Statement II: Freshly prepared starch solution is used as an indicator in iodometric titrations.
In the light of above statements, choose the correct answer from the options given below

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

59. Consider the following statements regarding orthoboric acid.
(A) It can be prepared by acidifying an aqueous solution of borax.
(B) It is sparingly soluble in water but highly soluble in hot water.
(C) It behaves as a weak acid in water due to self-ionization.
(D) Acidity of an aqueous solution of orthoboric acid increases upon addition of cis-diol.
Choose the correct statements.

(1) (A) and (B) only

(2) (A), (B) and (C) only

(3) (C) and (D) only

(4) (A), (B) and (D) only

CC-315 CC-315 CC-315 CC-315 CC-315 CC-315

60. Match the given List I and II and choose the correct option.

	List-I		List-II
(a)	Producer gas	(i)	$CO + H_2$
(b)	Synthesis gas	(ii)	SiO_2
(c)	Dry ice	(iii)	Solid CO_2
(d)	Silica	(iv)	$CO + N_2$

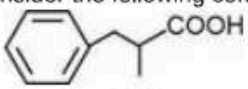
(1) a-(iv), b-(i), c-(iii), d-(ii)

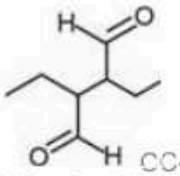
(2) a-(iv), b-(i), c-(ii), d-(iii)

(3) a-(i), b-(iv), c-(ii), d-(iii)

(4) a-(i), b-(iv), c-(iii), d-(ii)

61. Consider the following compounds and their IUPAC names

(a)  2-Methyl-3-phenyl propanoic acid

(b)  Hexan-3,4-dial

(c) $HC \equiv C - CH_2 - CH_2 - CH = CH_2$ Hex-1-en-5-yne

The correct IUPAC names of the compounds are given as

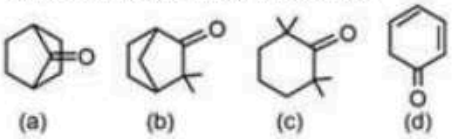
(1) (b) and (c) only

(2) (a) and (b) only

(3) (a), (b) and (c)

(4) (a) and (c) only

62. Consider the following compounds:



(a) (b) (c) (d)

Which among the given compounds can exhibit tautomerism?

(1) (a) and (d)

(2) (d) only

(3) (b) only

(4) (a), (b), (c) and (d)

63. In Duma's method for estimation of nitrogen, 0.5 g of an organic compound gave 50 mL of nitrogen collected at 300 K temperature and 720 mmHg pressure. If the aqueous tension at 300 K is 20 mmHg, the percentage of nitrogen in the compound is

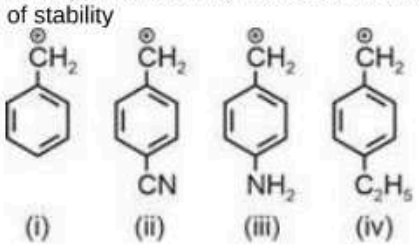
(1) 10.47%

(2) 25.27%

(3) 40.24%

(4) 74.47%

64. Arrange the following carbocations in their decreasing order of stability



(i) (ii) (iii) (iv)

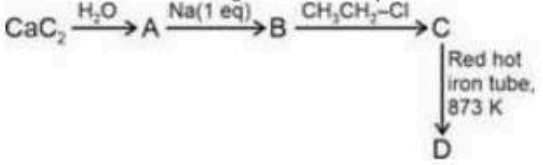
(1) (i) > (iii) > (iv) > (ii)

(2) (iii) > (iv) > (i) > (ii)

(3) (ii) > (i) > (iv) > (iii)

(4) (iv) > (iii) > (i) > (ii)

65. Consider the following reaction sequence



Major product D is

(1) Benzene

(2) Ethylbenzene

(3) Toluene

(4) 1,3,5-triethylbenzene

66. Match List-I (Salts with degree of dissociation α) with List-II (van't Hoff factor (i)).

List-I	List-II
a. NaCl (α = 90%)	(i) 3.25
b. $CaCl_2$ (α = 60%)	(ii) 2.2
c. $AlCl_3$ (α = 50%)	(iii) 1.9
d. $K_3[Fe(CN)_6]$ (α = 75%)	(iv) 2.5

The correct option is

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iii), b(ii), c(iv), d(i)

(3) a(ii), b(i), c(iii), d(iv)

(4) a(iv), b(iii), c(ii), d(i)

67. 5 moles of liquid X and 10 moles of liquid Y are mixed to form a solution having vapour pressure of 75 torr. The vapour pressures of pure X and Y are 63 torr and 78 torr respectively. Based on the given information consider the following statements:

(a) The solution shows positive deviation from Raoult's law.

(b) The X–Y interaction is greater than X–X and Y–Y interaction.

(c) Heat is absorbed during mixing.

The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

CC-315 CC-315 CC-315 CC-315 CC-315 CC-315

68. Volume of gases obtained at STP on electrolysis of $AgNO_3$(aq) using inert electrodes when 9.65 amperes current is passed for 60 minutes, is

(1) 2 L

(2) 4 L

(3) 8 L

(4) 10 L

69. If $E^{o}_{Cu^{2+}/Cu} = +0.34\,V$ then reduction potential of Cu^{2+}/Cu electrode at pH = 14 is (K_{sp} of $Cu(OH)_2 = 1 \times 10^{-19}$)

(1) –0.92 V

(2) – 0.22 V

(3) –0.65 V

(4) – 0.76 V

70. The half-life period of a first order reaction is 30 minutes. What fraction of the reactant remains after 70 minutes? (antilog 0.702 = 5.036)

(1) 0.5

(2) 0.2

(3) 0.7

(4) 0.6

71. Identify the correct orders against the property mentioned
(a) $H_2Te > H_2O > H_2S$(Melting point)
(b) $H_2Se > H_2S > H_2O$(Dissociation constant in aqueous medium)
(c) S > Se > Te (Negative electron gain enthalpy)
(d) Te > Po > Se (Melting point)
Choose the correct answer from the options given below:

(1) (a), (b) and (c) only

(2) (b), (c) and (d) only

(3) (b) and (c) only

(4) (a), (b), (c) and (d)

72. Consider the reaction,

$HNO_3 \xrightarrow{P_4O_{10}}$ compound(A) + HPO_3.

Incorrect statement about compound (A) is

(1) It has N–O–N bond

(2) Covalency of nitrogen is 5

(3) It is acidic in nature

(4) It is ionic in nature in solid state

73. The element of 3d, 4d & 5d series with highest melting point are respectively

(1) Mn, Mo, W

(2) Mn, Tc, W

(3) Cr, Tc, W

(4) Cr, Mo, W

74. $Cr_2O_7^{2-} \underset{pH = B}{\overset{pH = A}{\rightleftharpoons}} CrO_4^{2-}$; A and B can be respectively.

(1) 2, 10

(2) 8, 4

(3) 4, 8

(4) 8, 10

75. Out of the following, choose a correct statement?

(1) $[Cu(NH_3)_6]^{2+}$ is a colourless ion

(2) $[Zn(H_2O)_6]^{2+}$ ion is blue coloured

(3) $[Ni(CO)_4]$ ion has a square planar shape

(4) If $[CoCl(NH_3)_5]^{2+}$ absorbs yellow colour of light, then the colour of this coordination entity is violet.

76. How many isomers can exist for the complex $[Cr(en)_2Br_2]Cl$?

(1) 3

(2) 6

(3) 2

(4) 10

77. Which of the following metal carbonyls has longest M-C bond length?

(1) $[Mn(CO)_6]^+$

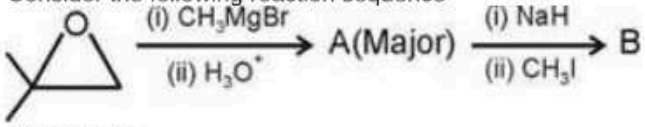
(2) $[Cr(CO)_6]$

(3) $[Fe(CO)_5]$

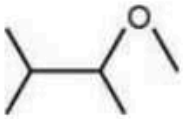
(4) $[V(CO)_6]^-$

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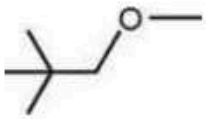
78. Consider the following reaction sequence

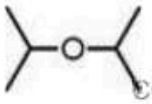


Product B is

(1) 

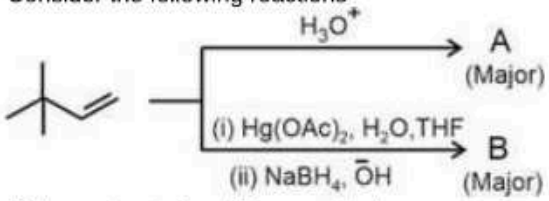
(2) 

(3) 

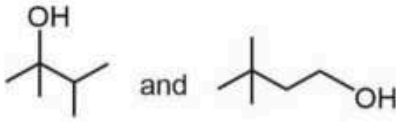
(4) 

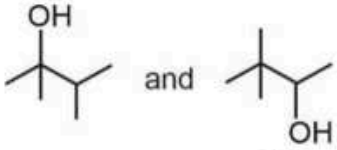
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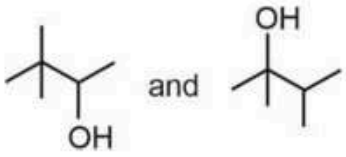
79. Consider the following reactions



Major products A and B respectively are

(1) 

(2) 

(3) 

(4) 

CC-315

80. Given below are two statements

Statement I: The correct order of bond enthalpies of 'C–X' bond is $CH_3–F > CH_3–Cl > CH_3–Br > CH_3–I$.

Statement II: The total number of possible structural isomers with molecular formula $C_5H_{11}Cl$ is seven.

In the light of above statements, choose the most appropriate answer from the options given below.

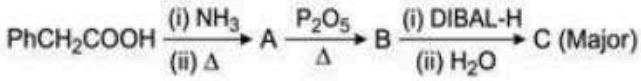
(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

81. Consider the given reaction sequence



Major product C is

(1) $PhCH_2CH_2NH_2$

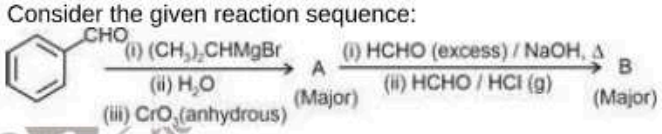
(2) $PhCH_2CHO$

(3) $PhCH_2CH_2OH$

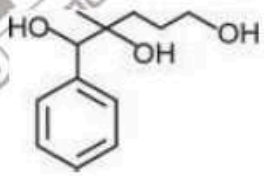
(4) $PhCH_2CH_3$

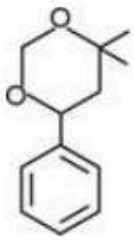
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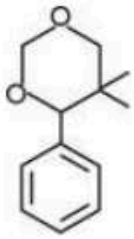
82. Consider the given reaction sequence:

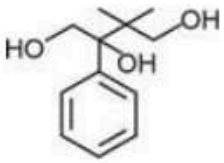


Major product B is

(1) 

(2) 

(3) 

(4) 

CC-315

83. Given below are the two statements: one is labelled as **Assertion (A)** and other is labelled as **Reason (R)**.
Assertion (A): 2-Methylpropanal undergoes aldol condensation reaction when heated with aqueous NaOH solution.
Reason (R): 2-Methylpropanal contains α-hydrogen.
In light of above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

84. Group reagents for Mn^{2+} and Sr^{2+} in qualitative analysis of cations respectively are

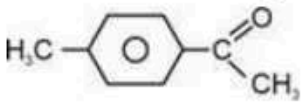
(1) H_2S in presence of dil. HCl and NH_4OH in presence of NH_4Cl

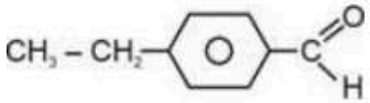
(2) Dilute HCl and $(NH_4)_2CO_3$ in presence of NH_4OH

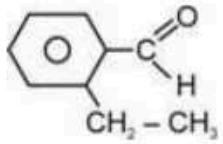
(3) H_2S in presence of NH_4OH and dilute HCl

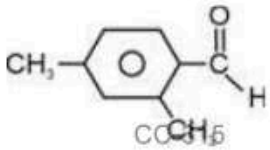
(4) H_2S in presence of NH_4OH and $(NH_4)_2CO_3$ in presence of NH_4OH

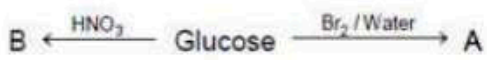
85. An organic compound with the molecular formula $C_9H_{10}O$ forms $2,4-DNP$ derivative, reduces Tollens' reagent and undergoes Cannizzaro reaction. On vigorous oxidation, it gives 1, 2-benzenedicarboxylic acid. The possible structure of the compound is

(1) 

(2) 

(3) 

(4) 

86. 

A and B respectively are

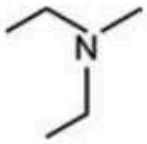
(1) Gluconic acid, Gluconic acid

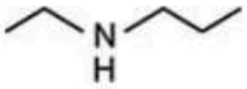
(2) Saccharic acid, Saccharic acid

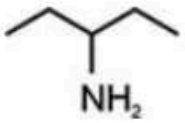
(3) Saccharic acid, Gluconic acid

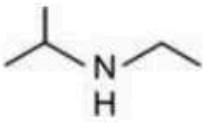
(4) Gluconic acid, Saccharic acid

87. A compound 'X' having molecular formula $C_5H_{13}N$ on reaction with Hinsberg's reagent gives compound 'Y' which dissolves in aqueous alkali. The structure of the compound 'X' could be

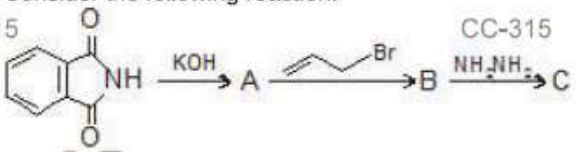
(1) 

(2) 

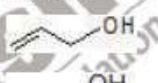
(3) 

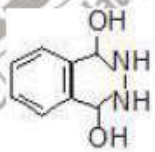
(4) 

88. Consider the following reaction.

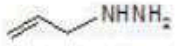


Product C is

(1) 

(2) 

(3) 

(4) 

89. Ammonium carbamate dissociates as $NH_2COONH_4(s) \rightleftharpoons 2NH_3(g) + CO_2(g)$ In a closed vessel containing ammonium carbamate in equilibrium, ammonia is added such that the partial pressure of ammonia at new equilibrium is $\frac{3}{2}$ times to the original total pressure. The ratio of total pressure now to the original pressure is

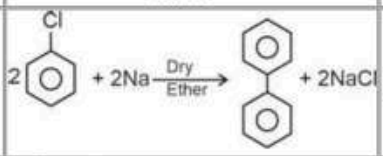
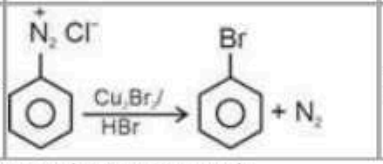
(1) $\frac{761}{486}$

(2) $\frac{486}{761}$

(3) $\frac{527}{364}$

(4) $\frac{672}{532}$

90. Match List I with List II

	List I (Chemical reaction)		List II (Name of the reaction)
a.	$CH_3Br + AgF \rightarrow CH_3F + AgBr$	(i)	Finkelstein reaction
b.	$CH_3Cl + NaI \xrightarrow[\text{acetone}]{\text{dry}} CH_3I + NaCl$	(ii)	Sandmeyer's reaction
c.		(iii)	Fittig reaction
d.		(iv)	Swarts reaction

Choose the correct match.

(1) a(iv), b(i), c(iii), d(ii)

(2) a(i), b(iv), c(iii), d(ii)

(3) a(i), b(iv), c(ii), d(iii)

(4) a(i), b(iii), c(ii), d(iv)

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BIOLOGY

91. In taxonomical hierarchy, higher the category
(a) Lesser is the difficulty of determining the relationship to other taxa at the same level
(b) Lesser is the number of common characteristics among the organisms
(c) More are the characteristics that the members within the taxon share
(d) Problem of classification becomes more complex
Choose the **correct** option.

(1) Only a and c are correct

(2) Only a, b and c are correct

(3) All a, b, c and d are correct

(4) Only b and d are correct

92. Invertase enzyme catalyzes the conversion of

(1) Sucrose into glucose and fructose

(2) 1, 3 bisphosphoglyceric acid into 3-phosphoglyceric acid

(3) Phosphoenol pyruvate into pyruvic acid

(4) Glyceraldehyde-3-phosphate into DHAP

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93. State true/false **(T/F)** for the statements given below and select the **correct** option.
a. Sexual spores are produced endogenously in *Claviceps*.
b. Dikaryophase is seen in the life cycle of *Mucor* and *Ustilago*.
c. Spores of slime moulds possess true walls and can survive for many years.
d. Methanogens are found in marshy areas and in the gut of several ruminants.
e. Biological classification of organisms as plants and animals was first proposed by Linnaeus.

(1) a-T; b-F; c-F; d-T; e-T

(2) a-T; b-F; c-T; d-T; e-F

(3) a-F; b-T; c-T; d-F; e-F

(4) a-F; b-F; c-T; d-T; e-F

94. Linnaeus artificial classification system of angiosperms was based on

(1) Stomata in leaves

(2) Internal structure of stem

(3) Modification of different parts of plants

(4) Androecium structure

95. Select the **incorrect** statement w.r.t. ETS and aerobic respiration?

(1) Oxygen is vital since it drives the whole process by removing hydrogen from the system

(2) Ubiquinone receives reducing equivalents via both complex-I and complex-II

(3) Complex-IV of the ETS refers to cytochrome bc_1 complex

(4) In eukaryotes aerobic respiration takes place within mitochondria

96. PEP is primary CO_2 acceptor in all, **except**

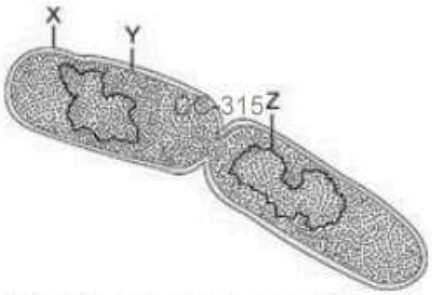
(1) *Sorghum*

(2) Maize

(3) Sugarcane

(4) Tomato

97.



Select the option that **correctly** depicts the above labels of the given organism.

(1) 'X' represents the cellulosic cell wall consisting of polysaccharide, along with amino acid

(2) 'Y' represents the capsule, which protects the organism from bursting and collapsing

(3) 'Z' represents the genetic material of the organism which is a circular DNA

(4) 'Y' represents the mucilaginous sheath that covers the cell wall of the organism

98. Female flowers of *Vallisneria* do **not** have

(1) Sticky stigma

(2) Unwettable stigma

(3) Nectaries

(4) Very long pedicel

99. Palisade parenchyma is absent in

(1) Leaves of potato

(2) Leaves of *Pisum sativum*

(3) Leaves of mango

(4) Leaves of *Triticum aestivum*

100. There are two leaves 'A' and 'B', which are increasing their area by showing growth in first, second and third year, respectively. Observe the given data of leaf area.

Leaf	Initial size	1st year size	2nd year size	3rd year size
'A'	15 cm^2	30 cm^2	60 cm^2	78 cm^2
'B'	30 cm^2	60 cm^2	75 cm^2	105 cm^2

Choose the **incorrect** one(s) from the following:

a. Leaf 'A' shows the same relative growth rate as that of leaf 'B' in the first year.

b. The absolute growth rate of leaf 'B' in first year and third year is same.

c. The relative growth rate of leaf 'A' is observed to be more than the relative growth rate of leaf 'B' in second year.

d. The relative growth rate of leaf 'B' in third year is double than that of leaf 'A' in third year

e. The absolute growth rate of leaf 'A' in second year is double than that of leaf 'B' in second year.

(1) c and d only

(2) a, c and e only

(3) d only

(4) b, d and e only

101. If a somatic cell has 40 pg of DNA in its G_2 phase, what will be the amount of DNA in the egg cell formed by it?

(1) 10 pg

(2) 20 pg

(3) 40 pg

(4) 80 pg

102. In a double stranded DNA, which of the given base ratios cannot necessarily be constant and equals one?

(1) $\frac{G}{C}$

(2) $\frac{A}{T}$

(3) $\frac{G+T}{A+C}$

(4) $\frac{G+C}{A+T}$

103. Root hairs are present in the region of

(1) Root cap

(2) Meristematic activity

(3) Elongation

(4) Maturation

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104. Observe the following food chain.
Grass → Grasshopper → Frog → Snake
If the energy of sunlight falling on grass is 1000 J, what will be the energy available to the frog?
Choose the **correct** option.

(1) 1 J

(2) 0.1 J

(3) 0.01 J

(4) 10 J

105. According to the five kingdom classification system by R.H. Whittaker in which pair of kingdoms, both autotrophic as well as heterotrophic mode of nutrition is found?

(1) Protista and fungi

(2) Fungi and Monera

(3) Animalia and Protista

(4) Monera and Protista

106. Match column I with column II.

Column I	**Column II**
a. Macrophage	(i) Releases of vasodilator
b. Mast cell	(ii) Synthesizes collagen fibres
c. Fibroblast	(iii) Performs phagocytosis
d. Adipocyte	(iv) Has peripheral nucleus

Choose the **correct** option

(1) a(i), b(ii), c(iv), d(iii)

(2) a(iii), b(i), c(ii), d(iv)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(ii), b(iii), c(iv), d(i)

107. Read the following statement and choose the option that **correctly** fills the blank.
The excess of nutrients which are not used immediately are converted into ____ and are stored in adipose tissue.

(1) Fats

(2) Proteins

(3) Carbohydrates

(4) Mucopolysaccharides

108. The structure of protein which contains only one polypeptide chain in it, can possess

(1) Only the structure in which haemoglobin exists for its biological activity

(2) Only the alpha helical secondary structure

(3) Only the structure in which positional information of a protein can be given

(4) Primary, secondary and tertiary structures

109. Competitive inhibitor of enzyme succinic dehydrogenase is

(1) Succinate

(2) Malonate

(3) Aspartic acid

(4) Alanine

110. In a nucleotide, an ester bond is present between the

(1) Sulphate and carboxyl group of sugar

(2) Phosphate and hydroxyl group of sugar

(3) Carbonate and carboxyl group of sugar

(4) Carboxyl and phosphate group of sugar

111. α-amino acid is a substituted

(1) Propane

(2) Methane

(3) Glycerol

(4) Ribose

112. **Assertion (A):** Different groups of animals respire through different respiratory structures like gills, lungs, tracheal tubes, *etc.*
Reason (R): Mechanism of breathing vary among different groups of animals depending only on their habitats.
In the light of above statements, select the correct option

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are false

(3) (A) is true but (R) is false

(4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

113. Select the name of the enzyme which facilitates the following reaction in RBCs.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$

(1) Carbonic anhydrase

(2) Dehydrogenase

(3) Hexokinase

(4) Transferase

114. Function not performed by conducting part of respiratory tract is

(1) Humidification of inhaled air

(2) Trapping of foreign particles

(3) Diffusion of O_2 and CO_2 between blood and tissues

(4) Bringing the inhaled air to the body temperature

115. All of the following animals circulate water from their surroundings through their body cavities to facilitate the cells to exchange essential substances, **except**

(1) *Sycon*

(2) *Hydra*

(3) *Pheretima*

(4) *Obelia*

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116. Read the following statements (a-d)
(a) These organisms can produce two kinds of spores.
(b) Sporophylls may form distinct, compact structures called strobili or cones.
(c) Fusion of male gamete with the egg present in the archegonium result in the formation of zygote.
(d) The female gametophytes are retained on the parent sporophytes permanently.
Select the features that are common in *Salvinia* and *Cycas*.

(1) Only (a) and (b)

(2) Only (a) and (c)

(3) Only (c) and (d)

(4) Only (b) and (c)

117. Read the following statements and select the **correct** option.
Statement A: Tropical Amazonian rain forest in South America has the greatest biodiversity on earth.
Statement B: As we move from low to high latitude, biodiversity increases.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

118. Which one of the following is an example of *in-situ* conservation?

(1) National park

(2) Zoological park

(3) Botanical garden

(4) Wildlife safari park

119. What is the net primary productivity of an ecosystem if the rate of production of biomass by solar energy is equal to the rate of respiration in producers?
Choose the **correct** option.

(1) Zero

(2) More than 1

(3) 1

(4) More than GPP

120. Which of the following statements does **not** represent the accurate reason for not copying both the DNA strands during transcription?

(1) One segment of the DNA would be coding for two different proteins

(2) The two RNA molecules if produced simultaneously would be complimentary to each other

(3) Formation of dsRNA will prevent translation

(4) Sequence of amino acids coded by both RNA molecules will be same

121. Read below given features
a. Stored food is very similar to amylopectin and glycogen in structure.
b. Complex post fertilization development occurs.
c. Absence of motile stage throughout their life cycle.
d. Deepest dweller.
Above given features are true for members of

(1) Chlorophyceae

(2) Phaeophyceae

(3) Rhodophyceae

(4) Chrysophytes

122. Regarding bulliform cells, select the option which is **odd** one out.

(1) These cell are found in the leaves of grasses

(2) These are colourless cells

(3) These are found in groups

(4) These are found on abaxial surface of leaves

123. All of the following are the functions of gibberellins, **except**

(1) GA_3 is used to speed up the malting process in brewing industry

(2) Spraying sugarcane crop with gibberellins increases the length of the stem

(3) Gibberellins promotes bolting just prior to flowering in beet and cabbages

(4) Inhibition of the activity of amylase enzyme and thus inducing the dormancy in seeds

124. The technology of biogas production was developed in India mainly due to efforts of

(1) IARI only

(2) People of rural areas only

(3) KVIC only

(4) Both IARI and KVIC

125. Given below are two statements labelled as Assertion (A) and Reason (R).
Assertion (A): The middle layer of the cell envelope of the cells that have fimbriae, prevent these cells from bursting or collapsing.
Reason (R): The middle layer of the cell envelope in the cells that shows the presence of fimbriae, has phospholipid as major component and differs in thickness among different forms.
In the light of the above statements, select the **correct** option:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

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126. A is the smallest living cell (0.3 µm long) and it lacks B.
Choose the option to fill in the blanks **correctly**.

(1) (A)- PPLO; (B)- Plasma membrane

(2) (A)- Mycoplasma; (B)- Cell wall

(3) (A)- *Amoeba*; (B)- Protein coat

(4) (A)- Bacterium; (B)- Ribosomes

127. Observe the following steps w.r.t. process, package and transport of materials by endomembrane system.
a. Movement of biomolecules from ER to Golgi apparatus in the form of transitional vesicles.
b. Release of vesicles from concave face of Golgi apparatus.
c. Synthesis of biomolecules in the network of tiny tubular structures scattered in the cytoplasm.
d. Movement of vesicle from forming to maturing face.
Select the option that depicts the correct chronological sequence of the given events.

(1) a → b → d → c

(2) c → a → d → b

(3) b → c → a → d

(4) c → b → d → a

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128. Read the following statements and select the **correct** option.
Statement A : In general, herbivores and plants appear to be more adversely affected by competition than carnivores.
Statement B : Predators are 'Prudent' in nature.

(1) Only statement A is correct

(2) Only statement B is correct

(3) Both statements are incorrect

(4) Both statements are correct

129. K in Verhulst-pearl logistic growth represents

(1) Population density

(2) Carrying capacity

(3) Reproduction rate

(4) Biotic potential

130. How many of the given features are **true** for the family to which *Datura* belongs?
(a) Tepals showing valvate aestivation.
(b) Endospermous seeds and fruits are berry or capsule.
(c) Type of inflorescence in which, main axis terminates into a flower.
(d) Leaves having alternate phyllotaxy and exstipulate condition.
(e) Stem is herbaceous, rarely woody.
Select the **correct** answer from the following options.

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(1) 5

(2) 3

(3) 4

(4) 2

131. Complete the analogy w.r.t. disorders of human circulatory system.
Angina pectoris : Acute chest pain :: Heart failure : _____
Choose the **correct** option.

(1) Atherosclerosis

(2) Sudden damage of heart muscle

(3) Reduced cardiac output

(4) Stoppage of the heart beat

132. Match the parts of nephron given in column I with their functions given in column II.

	Column I		**Column II**
a.	Proximal convoluted tubule	(i)	Reabsorption of HCO_3^-
b.	Descending limb of loop of Henle	(ii)	Reabsorption of essential nutrients
c.	Ascending limb of loop of Henle	(iii)	Permeable to water but almost impermeable to electrolytes
d.	Distal convoluted tubule	(iv)	Impermeable to water but allow transport of electrolytes

Select the **correct** option.

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(1) a(ii); b(iii); c(iv); d(i)

(2) a(i); b(iii); c(ii); d(iv)

(3) a(ii); b(iv); c(i); d(iii)

(4) a(iii); b(i); c(ii); d(iv)

133. **Assertion (A):** ADH can increase the blood pressure and thus can increase GFR.
Reason (R) : Its dilatory effect on blood vessels can never affect the kidney functions.
In the light of above statements, select the correct option.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct, (R) is incorrect.

(4) Both (A) and (R) are incorrect.

134. Calculate and select the total number of bones present in appendicular skeleton of humans.

(1) 206

(2) 126

(3) 80

(4) 30

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135. Select the **incorrect** statement.

(1) Thigh muscle fibres are striated and voluntary
(2) Iris is composed of involuntary smooth muscle fibres
(3) Biceps of upper arm are striped muscle fibres
(4) Heart wall is constituted by involuntary and unstriated muscle fibres

(1) (1)

(2) (2)

(3) (3)

(4) (4)

136. How many bones given in the box are unpaired?

Zygomatic, Hyoid, Sternum, Clavicle, Frontal, Parietal, Mandible, Sphenoid, Ethmoid, Occipital

Choose the **correct** option

(1) 5

(2) 6

(3) 7

(4) 8

137. Hyposecretion of which hormone since pregnancy causes stunted growth in humans?

(1) PRH

(2) FSH

(3) Thyroxine

(4) Insulin

138. Hormone that regulates gene expression by interaction with genome through the hormone receptor complex formation is

(1) TCT

(2) Estrogen

(3) Norepinephrine

(4) Oxytocin

139. How many of the following animals given below in the box are economically beneficial?

Apis, *Bombyx*, *Culex*, *Locusta*, *Pinctada*

Choose the correct option

(1) Two

(2) Three

(3) Four

(4) One

140. Match the following columns and select the **correct** option

	Column-I		Column-II
a.	Metagenesis	(i)	*Pleurobrachia*
b.	Eight ciliated comb plates	(ii)	*Limulus*
c.	Book gills	(iii)	*Obelia*
d.	Lac insect	(iv)	*Laccifer*

(1) a – (iii), b – (i), c – (ii), d – (iv)

(2) a – (iii), b – (ii), c – (i), d – (iv)

(3) a – (i), b – (ii), c – (iii), d – (iv)

(4) a – (iv), b – (i), c – (ii), d – (iii)

141. The digestive tract of birds have additional chambers A and B. Select the option which fill the blanks **correctly**.

(1) Colon, caecum

(2) Jejunum, crop

(3) Crop, gizzard

(4) Air cavities, crop

142. What is the function of the structure labelled as 'X'?



(1) Removal of excretory products from haemolymph

(2) Secretion of digestive juices

(3) Absorption of digested food

(4) Grinding of food particles

143. Select the **correct** statement w.r.t. cockroach.

(1) The body of cockroach is divided into cephalothorax and abdomen.

(2) In each segment, endoskeleton comprises hardened plates called sclerites.

(3) Sclerites of each segment are joined together by articular membrane.

(4) Head is formed by the fusion of six embryonic segments and it is immovable due to fixed neck.

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144. Adult *Rana tigrina* can respire *via*
(a) Skin
(b) Buccal cavity
(c) Gills
(d) Lungs
Select the **correct** option.

(1) a, b and c

(2) b, c and d

(3) a, c and d

(4) a, b and d

145. Milk can be temporarily stored in which part of mammary gland?

(1) Mammary duct

(2) Mammary alveoli

(3) Lactiferous duct

(4) Areola

146. Choose the **incorrect** match w.r.t. menstrual cycle.

(1) Ovulation - LH surge

(2) Secretory phase- - Development of corpus luteum and increased secretion of progesterone

(3) Menstrual phase - Breakdown of perimetrium when ovum is fertilized

(4) Proliferative phase - Rapid regeneration of endometrium

147. The zygote or early embryos upto 8 blastomeres is transferred into the __A__ and this is called __B__. Here A and B are

(1) **A** – Fallopian tube, **B** – GIFT

(2) **A** – Uterus, **B** – IUI

(3) **A** – Uterus, **B** – IUT

(4) **A** – Fallopian tube, **B** – ZIFT

148. Which of the following is **correct** w.r.t. vasectomy?

(1) A small part of fallopian tube is removed

(2) Transport of ovum is blocked

(3) Testosterone production stops

(4) Libido remains unaffected

149. All of the following statements are considered as connotations of theory of special creation **except**

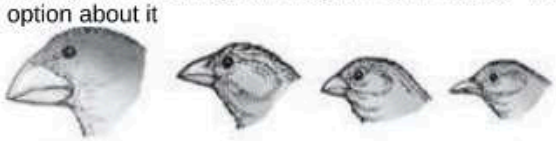
(1) Earth is about 4000 years old

(2) Diversity was always the same since creation

(3) All living organisms were created as such

(4) Life comes only from pre-existing life

150. Identify the below given diagram and choose the **correct** option about it



(1) Shows variety of beaks of Darwin's finches

(2) Shows adaptive convergence

(3) Shows coevolution

(4) Shows convergent evolution

151. Select the **incorrect** statement w.r.t. malaria.

(1) The malarial parasite requires two hosts – human and female *Anopheles* mosquito.

(2) *Plasmodium* reproduces asexually in the liver cells of humans.

(3) The male gamete and female gamete of *Plasmodium* fuse in the mosquito's gut.

(4) Sporozoites escape from the mosquito's gut and migrate to the mosquito's sweat glands.

152. Complete the analogy w.r.t. the disease and its causative agent.
Typhoid : *Salmonella typhi* : : Pneumonia : _____

(1) *Rhino* virus

(2) *Streptococcus pneumoniae*

(3) *Plasmodium vivax*

(4) *Entamoeba histolytica*

153. Who among the following had developed a method for removing plasmids from the cell and reinserting them in other cells?

(1) Stanley Cohen

(2) Herbert Boyer

(3) Fredrick Sanger

(4) Paul Ehrlich

154. The method by which recombinant DNA can be precisely incorporated directly into the nucleus of an animal cell is

(1) Microinjection

(2) Biolistics

(3) Gene gun

(4) Electroporation

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155. Consider the given statements and choose the incorrect one.

(1) By the end of 12 weeks of pregnancy, the limbs and external genital organs are well-developed.

(2) By the end of about 24 weeks of pregnancy, the body of foetus is covered by fine hair.

(3) By the end of 10 weeks of pregnancy, eye-lids separate and eyelashes are formed.

(4) After one month of pregnancy, the embryo's heart is formed.

156. Match List-I with List-II and select the **correct** option.

	List-I		List-II
a.	Chlorophyll *b*	(i)	Protects the chief pigment from photo- oxidation
b.	Xanthophyll	(ii)	It is the chief photosynthetic pigment
c.	Chlorophyll *a*	(iii)	It appears yellow in the chromatogram
d.	Carotenoid	(iv)	It appears yellow-green in chromatogram

(1) a(iv), b(i), c(iii), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(i), c(iv), d(iii)

157. Formation of interzonal fibres occur in

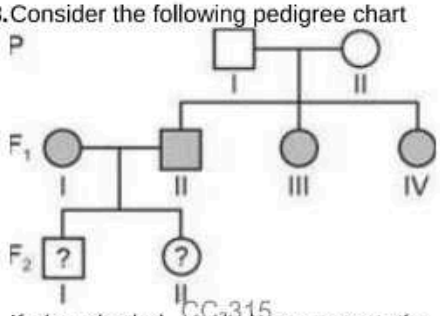
(1) Anaphase

(2) Metaphase

(3) Telophase

(4) Prophase

158. Consider the following pedigree chart



If the shaded symbols represent the affected ones then which of the following statements should be **true**?

(1) The trait is due to X-linked recessive gene

(2) The individual P (I) is homozygous for the related gene

(3) The trait cannot be a Mendelian disorder

(4) The individuals F_2 (I) and F_2 (II) must be affected

159. Identify the **incorrect** statement.

(1) The hilum is a scar on the seed coat through which the developing seeds were attached to the fruit.

(2) In coconut, the mesocarp is fibrous.

(3) The outer covering of endosperm in maize seed separates the embryo by proteinaceous layer called aleurone layer.

(4) The plumule and radicle are enclosed in sheaths which are called coleorhiza and coleoptile respectively.

160. Match the following columns and select the **correct** option.

Column I	Column II
a. Microtubules	(i) Help in pseudopodia formation
b. Microfilaments	(ii) Help in the formation of basal bodies
c. Centrioles	(iii) Involved in formation of scaffold for chromatin
d. Intermediate filaments	(iv) Form cytoskeleton of cilia and flagella

(1) a(ii), b(iv), c(i), d(iii)

(2) a(iv), b(ii), c(iii), d(i)

(3) a(iv), b(i), c(ii), d(iii)

(4) a(iii), b(ii), c(iv), d(i)

161. ABO blood group is a good example of

(1) Incomplete dominance

(2) Complete Linkage

(3) Lethality

(4) Multiple alleles

162. Arrange the following events/stages occuring during meiosis in the **correct** sequential order.

a. Univalent chromosomes start moving towards opposite poles.

b. Synaptonemal complex starts to dissolve.

c. Chromatids reach the opposite poles and start to uncoil.

d. Recombination between homologous chromosomes is completed and the two chromatids are linked at the site of crossing over.

e. The centriole duplicate in the animal cells.

Choose the **correct** answer from the given options.

(1) d → b → a → e → c

(2) b → d → a → e → c

(3) b → d → c → e → a

(4) e → d → b → c → a

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163. Which of the following statements is **incorrect**?

(1) Synergids secrete some chemotropic substance to direct the growth of pollen tube

(2) The most common type of embryo sac in angiosperm is monosporic which is formed by three sequential mitotic divisions.

(3) Antipodal cells are the vegetative cells of embryo sac

(4) In angiosperms, the egg apparatus of embryo sac consists of one egg cell , two synergids, three antipodal cells and two polar nuclei

164. Read the following **Assertion (A)** and **Reason (R)** statements and choose the **correct** answer.
Assertion (A): In many *Citrus* varieties, each ovule contains multiple embryos.
Reason (R): In *Citrus*, some of the nucellar cells surrounding the embryo sac start dividing, protrude into the embryo sac and develop into the embryos.

(1) Both (A) and (R) are false

(2) Both (A) are (R) are true but (R) is not the correct explanation of (A)

(3) Both (A) are (R) are true and (R) is the correct explanation of (A)

(4) (A) is true but (R) is false

165. Select the **mis-matched** pair.

(1) Cyclosporin A : *Trichoderma polysporum*

(2) Statins : *Monascus purpureus*

(3) Streptokinase : *Lactobacillus*

(4) Ethanol : *Saccharomyces cerevisiae*

166. *Lac z*, *lac y* and *lac a* gene in *Lac* operon are

(1) Structural genes

(2) Regulator genes

(3) Promoter genes

(4) Operator gene

167. The distance between genes A and B is 3 map units, between B and C is 10 map units and between C and A is 7 map units; the order of the genes in a linkage map constructed on the above data would perhaps be

(1) A, B, C

(2) A, C, B

(3) B, C, A

(4) B, A, C

168. Read the given assertion (A) and reason (R) and choose the **correct** option.
Assertion (A): During Griffith's experiment, the R-strain bacteria had somehow been transformed by heat-killed S-strain bacteria.
Reason (R): There might have been the transfer of some transforming principle to rough type bacteria from heat-killed smooth bacteria during Griffith's experiment, that had enabled R-strain to synthesise smooth polysaccharide coat.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

169. Match column I with column II.

	Column I		Column II
a.	Sickle-cell anaemia	(i)	Occurs due to the presence of defective blood clotting factor in affected individual
b.	α-Thalassemia	(ii)	Affected individual lacks liver enzyme phenylalanine hydroxylase
c.	Phenylketonuria	(iii)	Defect is caused by transversion mutation of the gene controlling β-chain of haemoglobin
d.	Haemophilia	(iv)	Controlled by two closely linked genes HBA1 and HBA2 on chromosome 16

Choose the **correct** option.

(1) a(iv), b(ii), c(i), d(iii)

(2) a(iii), b(iv), c(ii), d(i)

(3) a(iii), b(i), c(ii), d(iv)

(4) a(i), b(iii), c(iv), d(ii)

170. In an organism following stretch of a polypeptide is synthesized.
Histidine – Cysteine – Methionine – Tryptophan – Aspartic acid
Codons for the given amino acids are as follows:
His – CAU or CAC
Cys – UGU or UGC
Met – AUG
Trp – UGG
Asp – GAU or GAC
Here in the options only one strand of the double stranded DNA has been represented. Then which among the following can be sequence of the template in DNA that codes for the given peptide?

(1) 3' GTCCCACATGCAGTG 5'

(2) 5' GTCCCATACGCAGTG 3'

(3) 5' GTCCCACATGCAGTG 3'

(4) 3' GTCCATCCGGCAGTG 5'

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171. Which of the following methods of diagnosis is **not** suitable for early detection of a disease?

(1) ELISA

(2) PCR

(3) Urine analysis

(4) RDT

172. Read the following:
(a) Recognises a recognition sequence of five base pairs.
(b) Isolated from the bacteriophage that belongs to species '*Haemophilus*'.
(c) Cuts the DNA helix in their sugar-phosphate backbones.
Which among the above stated features is/are true w.r.t. *Hind* II?

(1) (a) and (b) only

(2) (b) and (c) only

(3) (c) only

(4) (a) only

173. Which of the following is **correct** related to mature insulin produced by biotechnology?

(1) It is less effective than natural insulin

(2) It is allergenic

(3) The main challenge for its production was getting it assembled into a mature form

(4) It shows different structure than natural one

174. Pia mater is **not**

(1) The innermost cranial meninx

(2) In contact with the brain tissue

(3) Highly vascular

(4) A tough thick, layer made up of fibrous connective tissue lacking blood vessels

175. A 28-year old woman has normal ovulatory cycles and a healthy uterus. Her partner produces adequate numbers of morphologically normal sperms but the sperms show poor motility and are unable to traverse the cervical mucus. If multiple attempts of conception have failed, then which assisted reproductive techniques would be most appropriate to achieve fertilization in this couple?

(1) GIFT

(2) AI

(3) ICSI

(4) IUI

176. The separated bands of DNA are cut from the agarose gel and extracted from the gel piece. This step is known as

(1) Electrophoresis

(2) PCR

(3) Elution

(4) Fragmentation

177. Which of the following is **not** a use of genetic modification in transgenic plants?

(1) Making crops more sensitive to abiotic stresses

(2) Reduced reliance on chemical pesticides

(3) Enhanced nutritional value of food

(4) Increased efficiency of mineral usage by plants

178. A normal adult human male ejaculates about ____ sperms during a coitus. Fill the blank with a suitable option.

(1) 100 to 200 million

(2) 100 to 200 billion

(3) 350 to 450 million

(4) 200 to 300 million

179. Through comparative investigation of vertebrate models, specifically humans and frogs, it was inferred that

(1) Both have ureters as the common urinogenital ducts

(2) Former have renal portal system unlike the latter

(3) Hindbrain in both of them consists of medulla, pons and cerebellum

(4) Latter have nucleated mature RBCs unlike the former which have enucleated mature RBCs

180. During the resting state, the axonal membrane of a neuron is

(1) Comparatively less permeable to K^+ than Na^+

(2) In a depolarised state

(3) Nearly impermeable to Na^+

(4) Permeable to negatively charged proteins

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